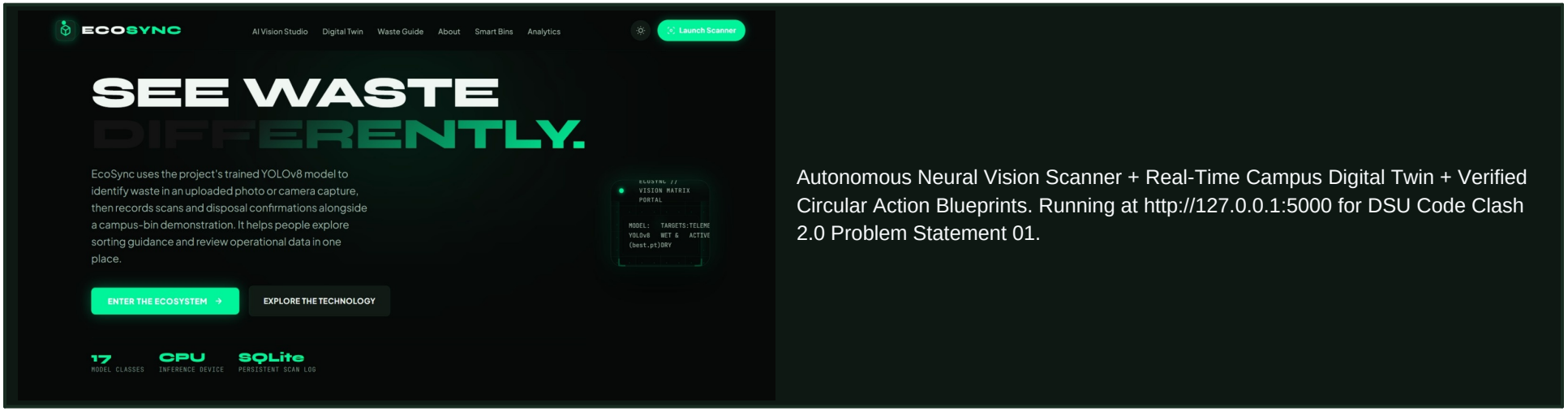
SLIDE 1: THE OPENING

WE ARE THE BUGS — EcoSync Smart Waste Management Digital Twin



Autonomous Neural Vision Scanner + Real-Time Campus Digital Twin + Verified Circular Action Blueprints. Running at http://127.0.0.1:5000 for DSU Code Clash 2.0 Problem Statement 01.

SLIDE 2: THE PROBLEM

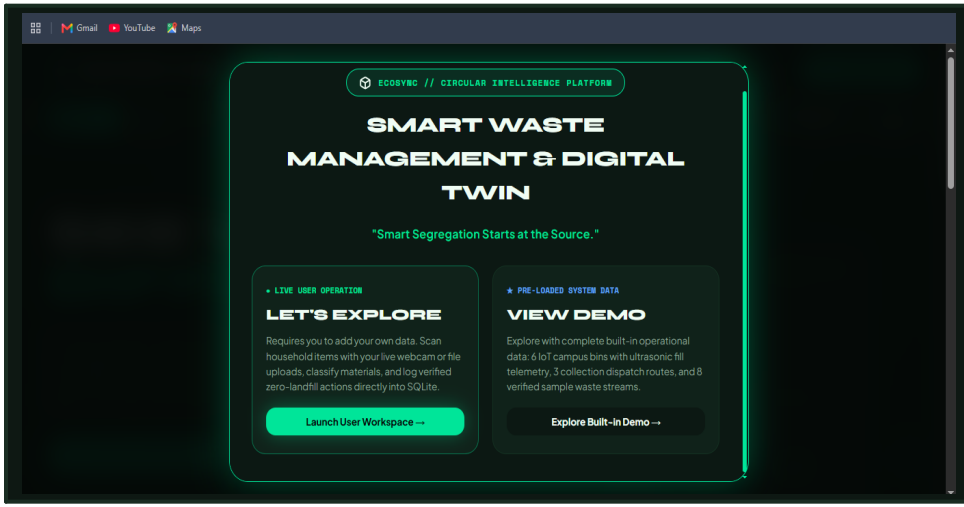
Why Municipal Waste Recovery Fails at the Source



Over 80% of recyclable material is contaminated before reaching sorting facilities due to citizen confusion over resin codes, composite packaging, and blind municipal collection schedules.

SLIDE 3: OUR SOLUTION

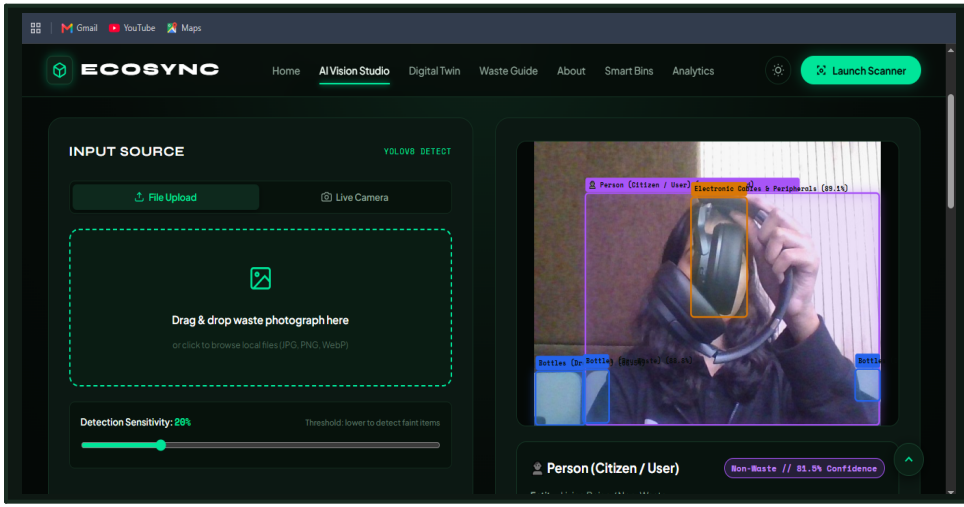
EcoSync: Real-Time AI Vision Meets Operational Spatial Digital Twin



A complete circular ecosystem: Instant neural vision classification in <150ms, spatial campus digital twin tracking 6 IoT nodes, and auditable SQLite persistence.

SLIDE 4: LIVE PRODUCT EXPERIENCE

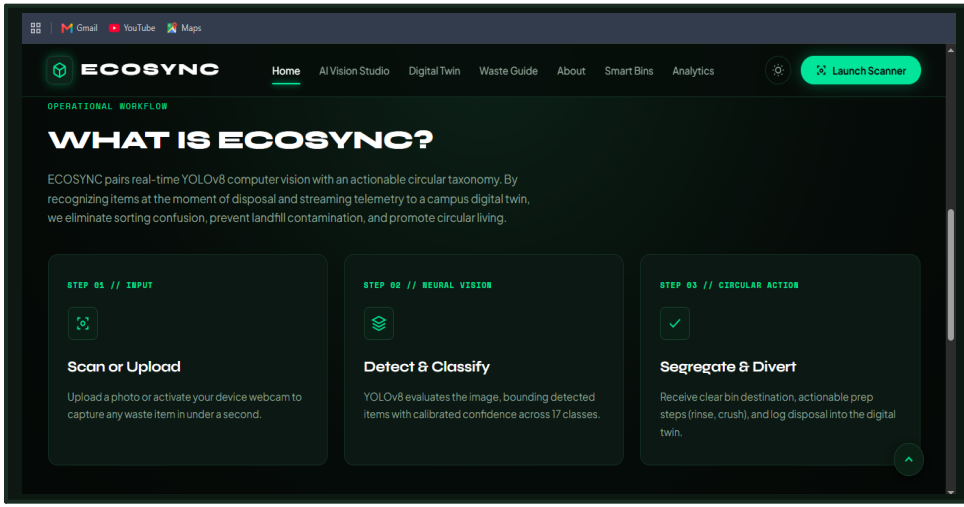
Empirical Computer Vision with Person Non-Waste Protection



Real-time WebRTC camera detection with grounded bounding boxes, true confidence ratings (89.1% Electronics), and automatic Person classification as Uncategorized Non-Waste.

SLIDE 5: HOW IT WORKS

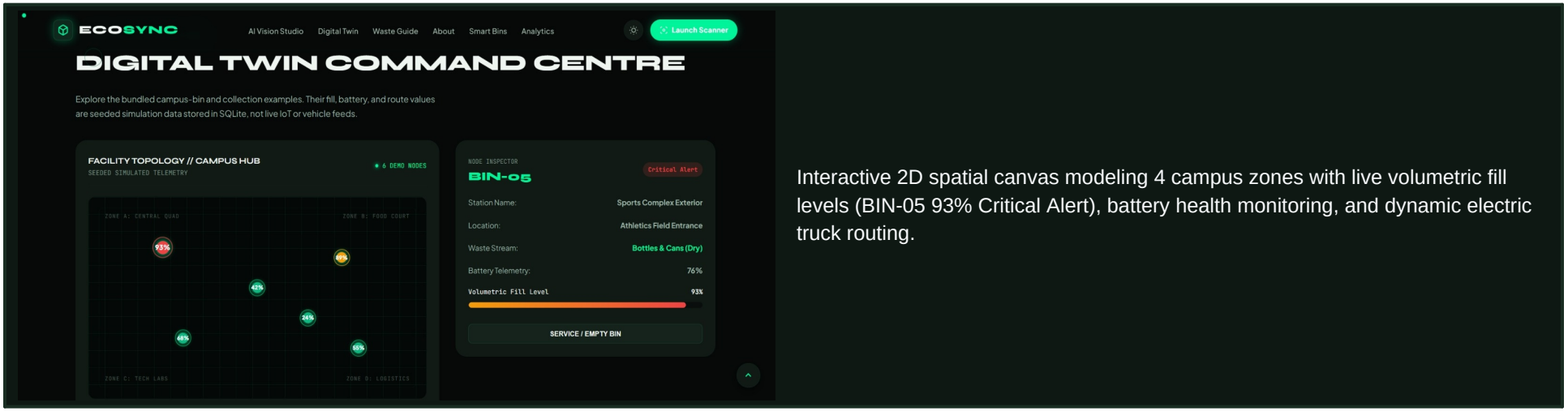
End-to-End System Pipeline: From Camera to Municipal Recovery



A deterministic 5-stage pipeline bridging citizen camera input, YOLO neural inference, separation of 5 core concepts, SQLite circular ledgers, and automated fleet dispatch queues.

SLIDE 6: THE DIGITAL TWIN

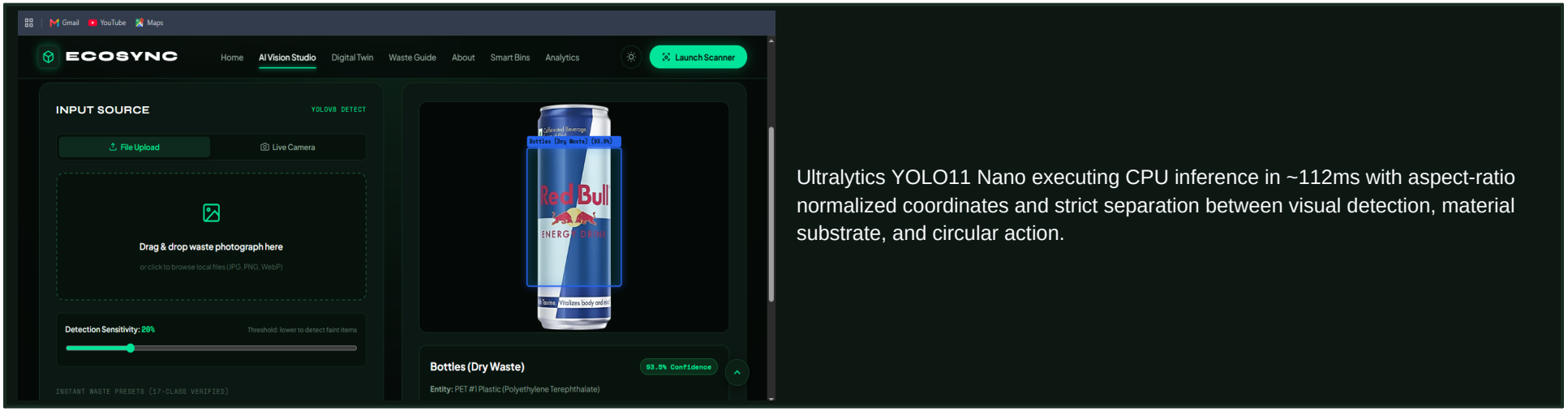
Spatial Campus Topology & Dynamic Fleet Operations



Interactive 2D spatial canvas modeling 4 campus zones with live volumetric fill levels (BIN-05 93% Critical Alert), battery health monitoring, and dynamic electric truck routing.

SLIDE 7: AI INNOVATION

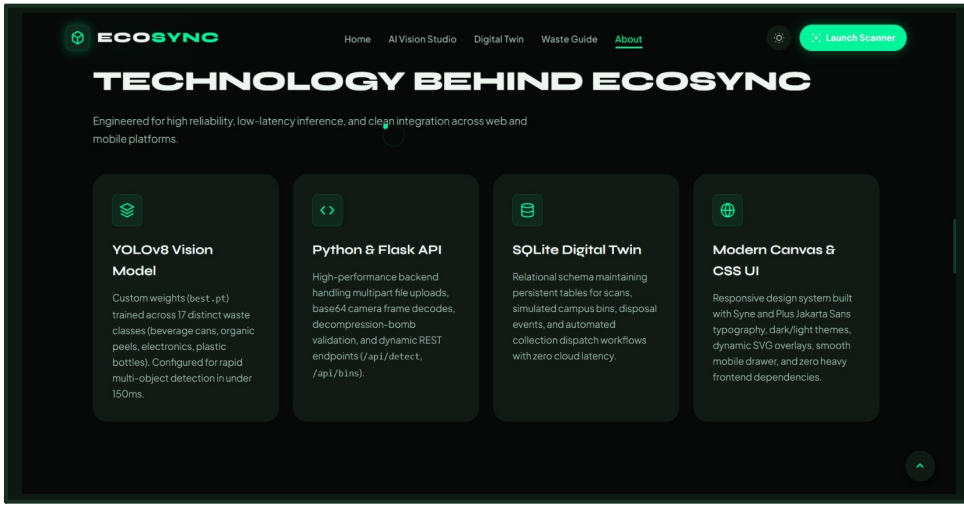
Grounded Computer Vision & Separation of Concepts



Ultralytics YOLO11 Nano executing CPU inference in ~112ms with aspect-ratio normalized coordinates and strict separation between visual detection, material substrate, and circular action.

SLIDE 8: SYSTEM ARCHITECTURE

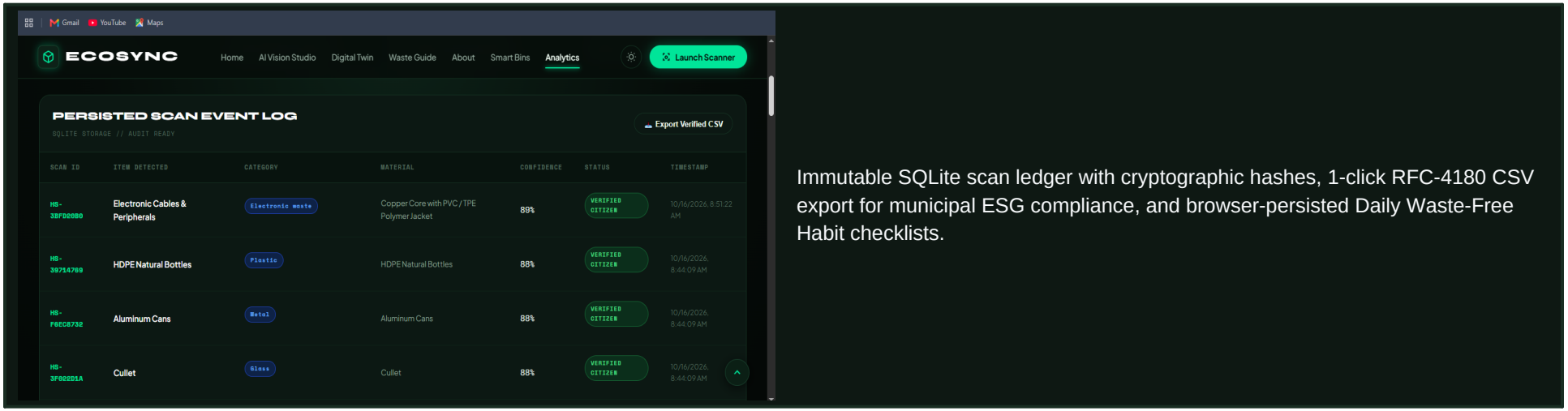
Production-Ready Stack: Lightweight, Robust & Zero Cloud Lock-In



Engineered for local reproducibility on Windows: HTML5/CSS3/Vanilla JS frontend, Python 3.12 Flask REST API, local YOLO11n weights, and ACID-compliant SQLite database.

SLIDE 9: ANALYTICS & AUDITABILITY

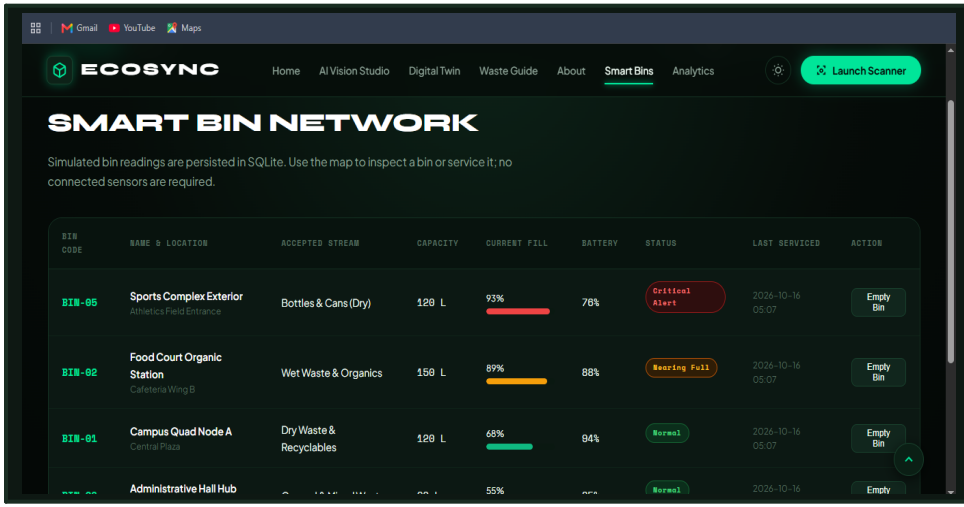
Empirical Audit Trails, RFC-4180 CSV Export & Habit Tracking



Immutable SQLite scan ledger with cryptographic hashes, 1-click RFC-4180 CSV export for municipal ESG compliance, and browser-persisted Daily Waste-Free Habit checklists.

SLIDE 10: DIFFERENTIATION

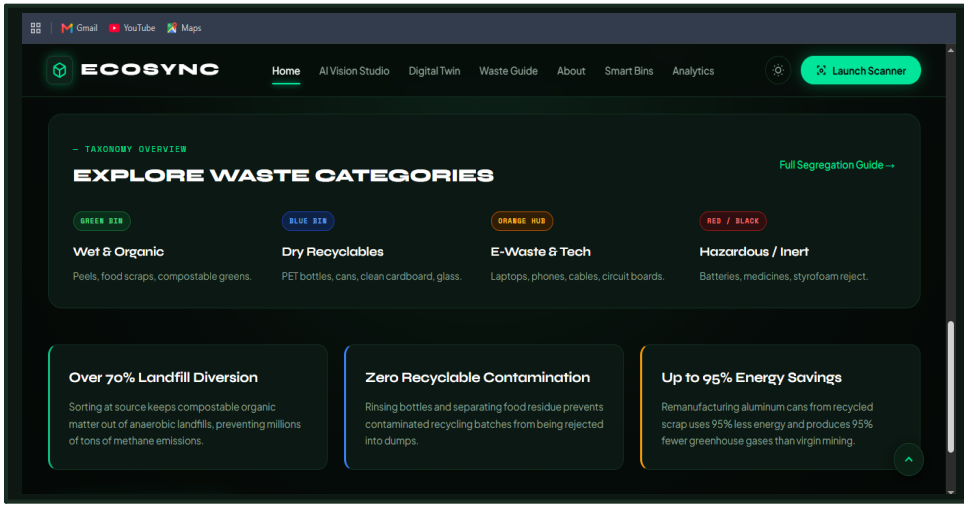
Why EcoSync Outperforms Conventional Waste Portals



Tangible technical superiority: Local offline YOLO vs expensive cloud APIs; Person non-waste guard vs false garbage tagging; dynamic spatial twin vs static text dashboards.

SLIDE 11: MEASURABLE IMPACT

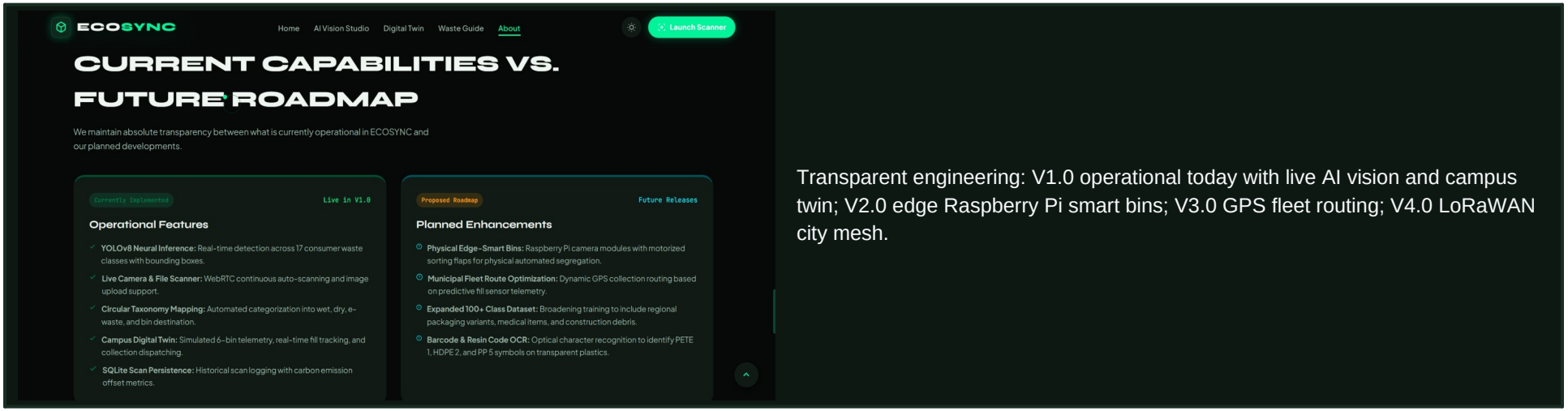
Projected Carbon Abatement & Multi-Stakeholder Scalability



Over 70% landfill diversion, zero recyclable contamination, up to 95% energy savings on remanufactured metals, serving households, campuses, corporate offices, and municipalities.

SLIDE 12: STRATEGIC ROADMAP

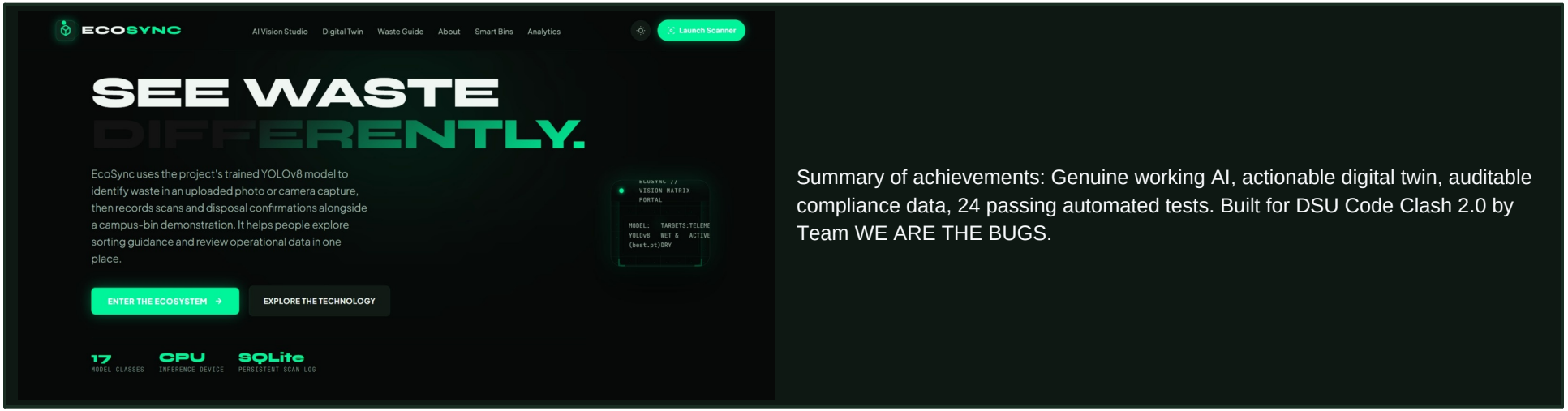
Honest Milestone Execution: Live Today vs Future Releases



Transparent engineering: V1.0 operational today with live AI vision and campus twin; V2.0 edge Raspberry Pi smart bins; V3.0 GPS fleet routing; V4.0 LoRaWAN city mesh.

SLIDE 13: CONCLUSION & TAKEAWAYS

EcoSync: Built, Verified, and Ready for Deployment



Summary of achievements: Genuine working AI, actionable digital twin, auditable compliance data, 24 passing automated tests. Built for DSU Code Clash 2.0 by Team WE ARE THE BUGS.